COMMERCIAL ITEM DESCRIPTION

FOUR CHANNEL RECORDER

The General Services Administration has authorized the use of this Commercial Item Description.

SCOPE. This Commercial Item Description (CID) specifies a portable four 50MM channel recorder which features a four-channel DC and amplifier range of 50MV to 500V full scale with a capability of a maximum common mode voltage of 500VDC. The recorder picks up test signals from a function generator that monitors three cylinder hydraulic fluid flows. The fourth channel monitors the changes between the three test signals. This system shall be known as a four channel recorder.

APPLICABLE DOCUMENTS.

MIL-M-7298	Manual Technical Commercial Equipment.
MIL-T-28800	Test Equipment for use with Electrical and Electronics Equipment General Specification for.
MIL-M-38797	Manual Technical, Operation Instructions and Maintenance Instructions (for Various Types of Equipment).

SALIENT CHARACTERISTICS.

1. Performance requirements of the four channel Oscillographic Recorder.

   a. Four channel recorder. The four channel recorder monitors four input test signals (5 volt full scale) with a 115VAC line voltage at a frequency range of 50 to 400HZ.

   b. Precision. The servo control penmoter should enforce 99.65% linearity over the entire channel, regardless of the channel width. Chart speed accuracy is + 0.25% from 15 to 35 degrees celsius temperature range and is independent of line frequency from 50 to 400HZ.

   c. Chart speed control. Must permit fixed or continuous variable operation of chart speed from 100% to 40% of selected speeds. The variable control is for reducing paper consumption or improving recording readability. Selection of the six calibrated speeds are by front-panel pushbuttons, or remote via TTL, or contact closure. The speed selections are 5, 10, 25, 50, 100, or 200 mm/sec and a divide-by-100 speed to reduce chart speed to 1/100th of the value selected by puchbutton. The external speed selection is achieved by applying an external pulse with a frequency ranging from 0 to 20 KHZ which corresponds with the resulting chart speed of 0 to 250 mm per second. Using this frequency range, the incremental chart drive response to the pulse at 30 microsecons with 10 microsecons minimum duration.

FSC 6625

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d. Trace. The pen must record in true rectilinear, precise, instant drying, and of constant width of 0.01 inches nominal at any chart speed velocity. It must have sufficient ink supply for at least one year of typical use and have a chart length of 275 feet nominal. The pen must record in any position, be clog free, and start writing immediately even after remaining idle for long periods. The tracer can be either pressurized fluid or fine-line thermal marking. For fine-line thermal marking the recorder should be resistant to pressure, will not wear out over the life of the recorder, and will not prematurely burn out even in the most demanding applications.

e. Event markers. The left and right event marker pens will have coding information and recording events capability. One of the event markers should also mark time which is activated by a timer that generates a 40-MS output pulse at eight internally programmable repetition rates of 0.1, 1, 10, 1000 seconds, and 0.01, 0.1, 1 and 10 minutes.

f. Operating temperature. The operating temperature for the four channel recorder is $0^{\circ}C$ at $+50^{\circ}C$.

g. Storage temperature. The storage temperature for the four channel recorder is $-40^{\circ}C$ to $+70^{\circ}C$.

h. Humidity. The humidity for the recorder is 95% at $32^{\circ}C$ noncondensing.

2. Performance requirements of each recorder.

a. Frequency response. The frequency response at 50-MM amplitude is DC to 50 HZ + 2%. For alternate amplitudes, such as 100-MM amplitude, it is DC to 30 HZ + 2%, and for 40-MM amplitude it is DC to 60 HZ + 2%.

b. Rise time. The rise time at 10% to 90% full scale with less than 1% overshoot is 5 milliseconds at 50-MM amplitude. Alternate rise time are 8 milliseconds at 100-MM amplitude, and 4 milliseconds at 40-MM amplitude.

c. Pen position. The analog pen can be positioned anywhere on the channel and still have a continuously adjustable + 100% full scale.

d. Common mode rejection noise ratio. The rejection ratio at 1 KOHM inbalance across the input signal points is 80 DB for DC signals and 60 DB for 60 HZ signals.

3. Stability.

a. Zero-line instability (drift). The zero-line instability of the recorder after 15 minutes of warmup is $\pm$ 0.1% of full scale for 24 hours or running, $\pm$ 0.05% of full scale per degree celsius, and $\pm$0.05% of reading for $\pm$ 10% line voltage change.

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b. Gain instability. The gain instability of the recorder is ±0.1% of reading for 24 hours of running, ±0.05% of reading per degree celsius, and ±0.05% of reading for ±10% line voltage change.

4. Specify type and size.

a. Design and construction. Unless otherwise specified herein, the equipment shall comply with the Type III design and construction requirements specified in MIL-T-28800.

b. Solid state construction. The equipment shall be of solid-state modular miniaturized construction. Electron tubes shall not be used.

c. Receptacle and connector mounting. All AC/DC input connectors are provided on barrier strip for jumper wire assembly. The connectors shall comply with MIL-T-28800 3.4.1.5.2.

d. Signal adjustment. All output signals shall be externally adjustable 100 to 40%.

e. Accessibility. The equipment shall be constructed so subassemblies and chassis components can be removed without removing any other hardware subassemblies, printed circuit card or component.

f. Input power switch. The input power switch shall comply with Type III input switch requirements of MIL-T-28800, except that the switch shall break both sides of the power source, but not break the earth ground conductor circuit.

g. Fuses or circuit breakers. Fuses or circuit breakers shall comply with Type III requirements of MIL-T-28800. Input power line shall be equipped with fuses or circuit breakers.

5. Dimensions, weight and mechanical stability.

a. Weight. The unpacked weight of the power supply, including all accessories, shall be 105 pounds nominal.

b. Dimensions. The dimensions of the four-channel recorder accessories, installed in the Style F enclosure as referenced in MIL-T-2800 shall be 19.31 inches in width, 17.22 inches in height and 14.75 inches in depth.

CERTIFICATION. The contractor shall certify that the product offered meets the salient characteristics of this description, and that the product conforms to the producer's own drawings, specifications, standards and quality assurance practices. Bid samples from bidders will be tested in accordance with the salient characteristics of this Commercial Item Description at San Antonio ALC, Kelly AFB, TX 78241.

OPERATIONAL INSTRUCTIONS/MAINTENANCE MANUAL. The contractor shall prepare an operator's manual according to the provisions of MIL-M-38797 for all equipment delivered under terms of this contract.

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COMMERCIAL PUBLICATIONS. Commercial publications may be submitted to fulfill Air Force documentation requirements. These publications require technical order numbering and must be referenced in the system level publication. Such commercial data must be prepared in accordance with MIL-M-7298. Two copies of each commercial manual proposed shall be submitted for Air Force evaluation and approval. Commercial publications shall be reviewed by the Technical Order Management Office (TOMO) before final acceptance by the Air Force. If modifications are performed on Air Force equipment, the contractor shall provide updating manual changes to meet Air Force standards. All applicable reproduction rights for government use must accompany submittals.

INTENDED USE. The equipment shall be designed for use at Air Force Base and Depot Precision Measuring Equipment Laboratories.

REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practical.

PACKAGING, PACKING AND MARKING. The items shall be packaged in accordance with normal commercial practice and packed to assure acceptance by common carrier and provide product protection against loss and damage during multiple shipments, handling, and storage. The shipping container shall be in compliance with national motor freight classification and uniform freight classification. Shipping containers shall be marked as specified in the contract or order.

KNOWN ACCEPTABLE PRODUCTS.

Manufacturer	Model	NSN
Gould Electronics and Electrical Products	P/N 2107-8809-00	6625-01-124-4185
For competitive bidding		

Custodian:
Navy - EC
Air Force - 99

Review activity:
Army - CR
DLA - ES

Preparing activity:
Air Force - 82

(Project 6625-0657)

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